

Undecidable Proofs

$$A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ accepts } w \}$$

Fact: A_{TM} is undecidable.

$$HALT_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM \& } M \text{ halts on input } w \}$$

Theorem 5.1: $HALT_{TM}$ is undecidable

Proof: Suppose not. $HALT_{TM}$ has a decider R . (R is an oracle/genie). We will use R to show that A_{TM} is also decidable, leading to a contradiction.

"R" on input $\langle M, w \rangle$.
If M accepts w or rejects w ,
accept.
If M loops on w ,
reject.

Here is a decider S for A_{TM} .

$S =$ "On input $\langle M, w \rangle$:

- Run R on $\langle M, w \rangle$ to see if M halts or loops.

- If R rejects, it means M loops, so reject.

- If R accepts, it means M halts.

So, it is safe to simulate M on w .

If M accepts w , accept.

If M rejects w , reject."

S is a decider for A_{TM} , a contradiction. $\square$

$$REG_{TM} = \{ \langle M \rangle \mid M \text{ is a TM \& } L(M) \text{ is regular} \}$$

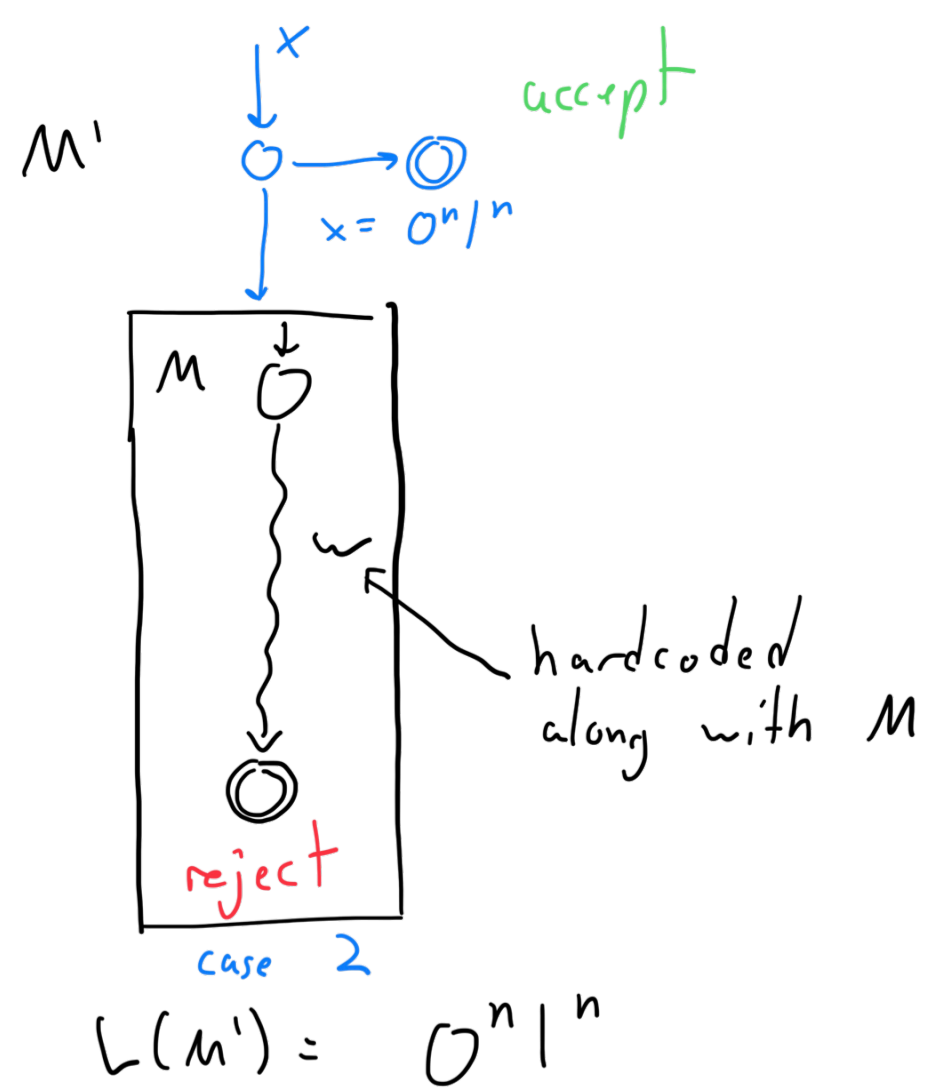
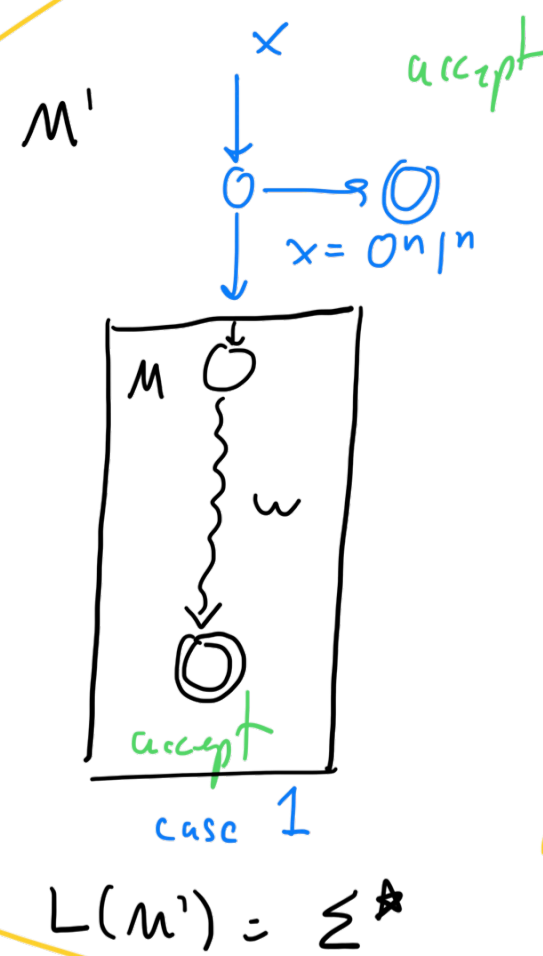
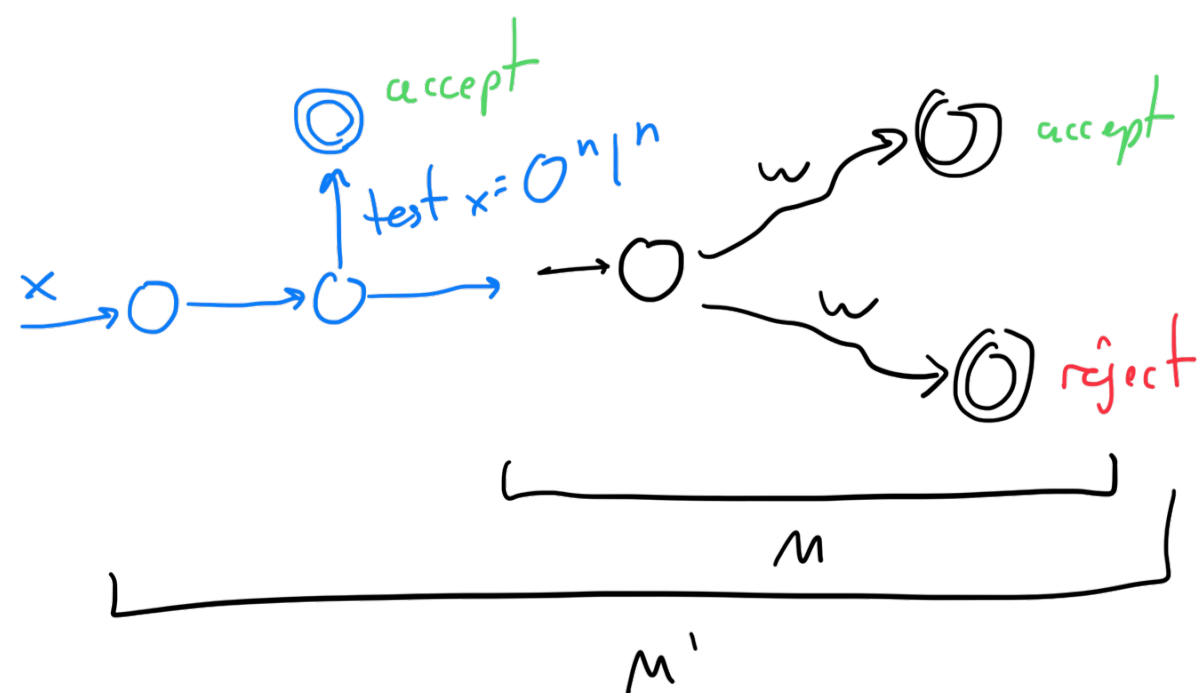
Theorem 5.3: REG_{TM} is undecidable.

Proof: Suppose not. It has a decider R .

We construct a TM to decide A_{TM} .

$S =$ "On input $\langle M, w \rangle$:

1. Modify M to create M' .



Don't run M' ! Just input M' into R to tell us if $L(M')$ is regular or not.

• If $L(M')$ is regular, then M accepts w , accept.

• If $L(M')$ is not regular, then M rejects w . reject."

S is a decider for A_{TM} , a contradiction. $\square$

$\square$